

A Multi-Stage Prompting and RAG Approach to Generating Legal Analysis in Common Law Systems

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Abstract. We explore the use of multi-stage prompting with retrieval-augmented generation (RAG) in generating legal analysis. We break the complex legal problems into small steps, including initial analysis, legal query formulation, legal sources retrieval, legal reasoning, result consolidation, and output generation. The design of this framework emulates the process a legal professional in a common law system approaches the problem.

Keywords. legal analysis, multi-stage prompting, retrieval-augmented generation

1. Introduction

Rising legal costs present a significant barrier to justice, prompting exploration of artificial intelligence as a potential solution. However, large language models (LLMs) often struggle with complex legal reasoning in zero-shot scenarios. This research investigates how a multi-stage prompting and retrieval-augmented generation (RAG) approach, designed to imitate important features of the cognitive process of common law lawyers, can enhance AI-generated legal analysis in common law systems, specifically focusing on the legal framework of the Hong Kong Special Administrative Region.

Legal analysis demands a structured approach, typically involving the identification of legal issues, searching for applicable rules from the sources of law (including legislation and precedents), and careful application of the rules. Our study proposes a novel five-stage pipeline meticulously crafted to emulate the thought process and methodology of experienced common law practitioners: legal issue identification, legal research question formulation, legal source retrieval, deductive legal reasoning, and legal reasoning by analogy. This methodology aims to break down the complex task of legal analysis into manageable steps for the machine, imitating the systematic approach employed by human lawyers and potentially improving the relevance, accuracy and reliability of AI-generated results.

This research aims to bridge the gap between expensive human legal expertise and the limitations of current AI systems in emulating human legal reasoning by closely

modeling important features of the reasoning process of common law lawyers. By doing so, we potentially pave the way for more accessible and reliable AI-assisted legal advice in common law jurisdictions, while enabling the answer generation processes of LLMs to imitate significant aspects of the legal analytical approach.

2. Method

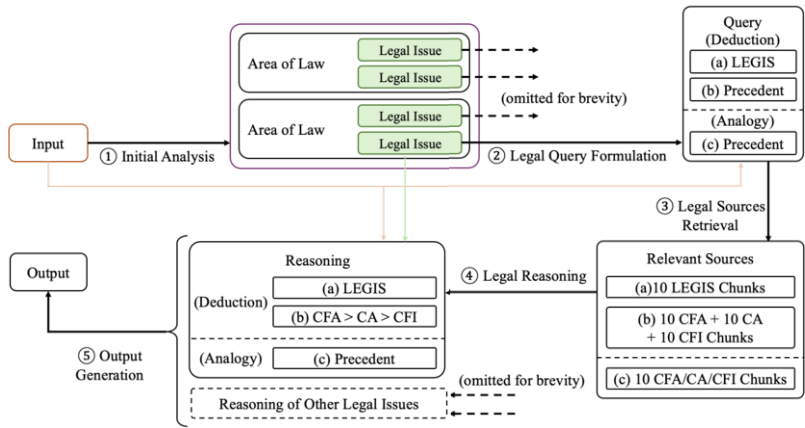


Figure 1. Framework of multi-stage prompting and RAG

Our multi-stage prompting system aims to emulate the reasoning process of a legal professional when answering a legal question. To achieve this goal, we break down the reasoning process into multiple steps, and design a module to handle each reasoning step. Figure 1 shows the diagram of our multi-stage prompting system. In this work, we adopt GPT-4o as the LLM for demonstration purpose. The prompt for identifying legal areas is omitted from the paper for brevity; the code, complete prompts, and examples can be found in <https://github.com/titusng0110/mirroring-legal-reasoning>.

Given an input question or scenario, a legal professional would first analyze the question to identify the areas of law and the issues pertinent to the question before conducting a deeper study. The initial analysis (Step ①) is mostly based on the knowledge possessed by the legal professional without collecting other legal materials. After identifying relevant areas of law and potential legal issues, the next step (②) is to formulate the legal queries for conducting searches in legal information systems. We emulate the process by prompting the LLM to generate queries. The queries will be used for retrieving (a) relevant legislation, (b) judgments related to the similar legal issue, and (c) judgments containing similar fact patterns (Step ③). The core of this study is to perform deductive and analogical reasoning in Step ④. Finally, the reasoning results would be concatenated and returned to the user (Step ⑤).

In this paper, we proposed an innovative idea of multi-stage prompting and RAG approach to generate legal reasoning. The prompt in each stage is meticulously crafted to instruct the LLM to focus on a small and specific task. Through this approach, we could inject argumentative knowledge such that LLM knows the procedure to follow. By capturing important features of human legal reasoning, this approach also potentially increase the explainability of LLM-generated results for legal problems.